



Topic: Recursion

Course: CSE 207 Data Structures
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Problems to Solve:

1. Design a recursive procedure to solve the Towers of Hanoi problem with n disks.
2. Write a recursive function to compute the sum of the digits of a given number.
3. Write a program to find the GCD of two numbers. You must use a recursive function to solve this problem.

Sample Input

- a. 18, 24
b. 13, 23

Sample Output

6
1

4. Suppose you are given a number n. You have to print n, n-1, n-2,.....1 on the console. Write a recursive function to solve the problem.

Sample Input

6

Sample Output

6 5 4 3 2 1

5. Write a program to calculate the summation of the first n elements of a Fibonacci sequence.
6. Write a program to delete the n-th node of a given linked list using recursion.

For example, a linked list

6	2	57	53	18
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After deleting the 3rd node, the list will be,

6	2	53	18
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7. Write a program to sort a linked list using recursion.

For example, a linked list:

61	26	59	31	18
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After sorting the list,

18	26	31	59	61
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